

HAT-P-12b

R-0657

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_260509 · created 2026-07-27T17:24:44+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461169.8291 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.016
Transit depth (Rp/Rs)^2	2.54 +/- 0.52 [%]
Orbital Inclination (inc)	89.86 +/- 0.93
Airmass coefficient 1 (a1)	1.167 +/- 0.021
Airmass coefficient 2 (a2)	-0.126 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.77 %
Best Comparison Star	None
Optimal Aperture	4.09
Optimal Annulus	5.98
Transit Duration (day)	0.0995 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.159 $\pm$ 0.016 vs 0.1406 (deviation 1.1 σ)
Duration fit vs published	0.0995 d vs 0.0975 d (deviation 0.77 σ)
Mid-time O-C	-2.3 min
Depth SNR	9.5
Residual scatter	1.77 %

Light curve

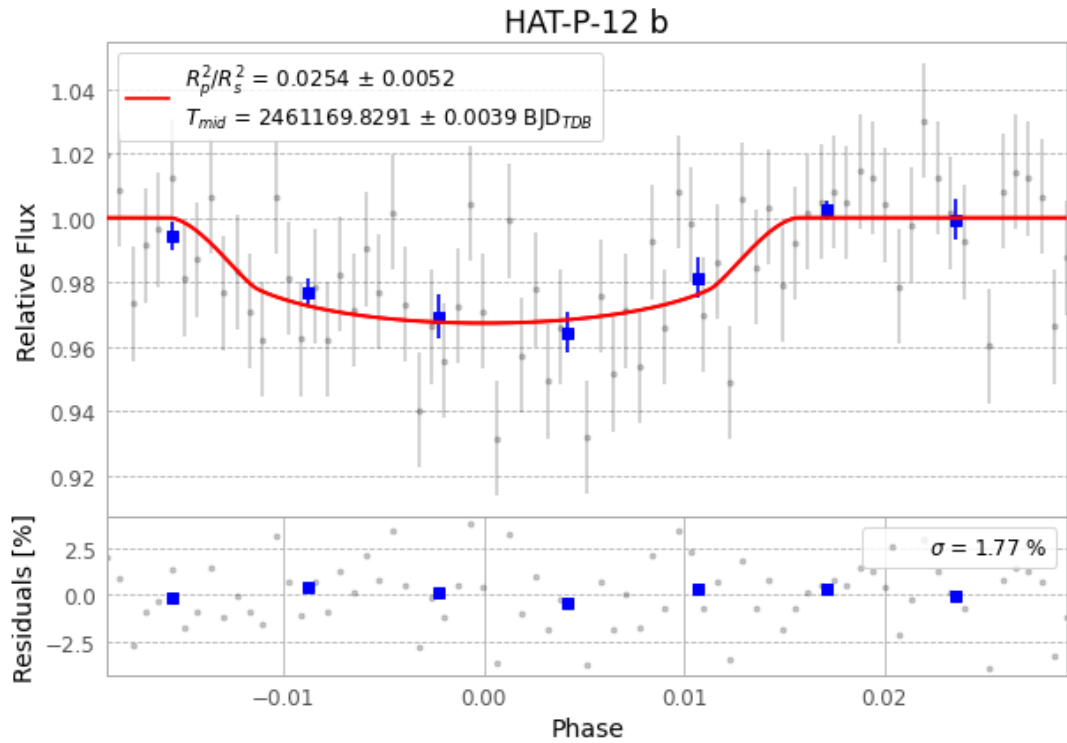


Figure 1 — FinalLightCurve_HAT-P-12 b_2026-05-08.png (SHA-256 f93e702b8649cb01...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [238, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f38b250606f74540...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e

Data provenance

75 FITS frames (49 MB), HATP-12260509062117.FITS → HATP-12260509100316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-260509.log (56d009a2ed19...), FinalLightCurve_HAT-P-12 b_2026-05-08.png (f93e702b8649...), FinalLightCurve_HAT-P-12 b_2026-05-08.csv (7f5413665d0f...)

Compute resources

Wall time 110.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 17:24Z.